

A 63-dimensional counterexample to Borsuk's conjecture

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Abstract

We give a finite construction showing that Borsuk's conjecture fails in dimension 63. Equivalently, if d_B denotes the least dimension in which Borsuk's conjecture fails, then

$$d_B \leq 63.$$

The construction is a 321-point set $X \subset \mathbb{R}^{63}$ such that every subset of strictly smaller diameter has at most 5 points. Hence any partition of X into smaller-diameter subsets requires at least $\lceil 321/5 \rceil = 65 > 64$ parts. The construction starts from the standard $G_2(4)$ representation used by Bondarenko and Jenrich–Brouwer; the new step is to take a 320-point subconfiguration in a codimension-two subspace and adjoin one projected point.

1 Introduction

Borsuk's conjecture asks whether every bounded set in $\mathbb{R}^d$ can be partitioned into $d+1$ subsets of strictly smaller diameter. Let d_B denote the least dimension in which the conjecture fails. Bondarenko's construction from the $G_2(4)$ graph gave $d_B \leq 65$ [1], and the Jenrich–Brouwer refinement gave $d_B \leq 64$ [2].

We prove the following improvement.

Theorem 1. *Borsuk's conjecture fails in dimension 63. In particular,*

$$d_B \leq 63.$$

The only external input is the standard $G_2(4)$ graph construction and its usual Euclidean representation. The additional finite facts about the partition used here are checked by the verification script in the accompanying repository [4]. From these facts we prove two new steps: a 320-point subconfiguration lies in a codimension-two subspace of the usual 65-dimensional representation, and one can adjoin a projected point in this subspace without allowing any smaller-diameter subset to have more than 5 points.

2 The finite model

We use Brouwer's projective model of the $G_2(4)$ graph [3]. Work in $\mathbb{F}_{16} = \mathbb{F}_2[\alpha]/(\alpha^4 + \alpha + 1)$ and in the projective plane $\text{PG}(2, 16)$. With respect to the Hermitian form

$$h(u, w) = u_0 w_0^4 + u_1 w_1^4 + u_2 w_2^4,$$

the vertices are the unordered triples

$$A = \{a_1, a_2, a_3\}$$

of pairwise orthogonal non-isotropic projective points. For such a triple A , let $T(A)$ be the associated isotropic 15-set, namely the set of isotropic projective points lying on the three projective lines spanned by pairs of points of A . Two vertices A, A' are adjacent when

$$|T(A) \cap T(A')| = 3.$$

Lemma 1 (finite verified facts). *The verification script in the accompanying repository [4] verifies the following facts.*

1. *The graph Γ has 416 vertices and is strongly regular with parameters*

$$(v, k, \lambda, \mu) = (416, 100, 36, 20).$$

Consequently, its nontrivial adjacency eigenvalues are 20 and -4 , with multiplicities 65 and 350, respectively.

2. *The clique number is*

$$\omega(\Gamma) = 5.$$

Equivalently, the script finds a 5-clique and verifies that there is no 6-clique.

3. *Fix the first isotropic point q_0 in the deterministic order used by the script. Let B be the set of vertices containing a non-isotropic point orthogonal to q_0 , and put $C = V(\Gamma) \setminus B$. Then*

$$|B| = 96, \quad |C| = 320.$$

The induced graph on B has three connected components

$$B_1, B_2, B_3$$

of size 32.

4. *The following degree data hold:*

$$\begin{aligned} |N(u) \cap B_i| &= 20 & (u \in B_i), \\ |N(u) \cap B_j| &= 0 & (u \in B_i, i \neq j), \\ |N(u) \cap C| &= 80 & (u \in B), \\ |N(c) \cap B_i| &= 8 & (c \in C, i = 1, 2, 3), \\ |N(c) \cap C| &= 76 & (c \in C). \end{aligned}$$

3 The standard Euclidean representation

Lemma 2 (standard representation). *There are vectors $x_v \in \mathbb{R}^{65}$, indexed by $v \in V(\Gamma)$, such that*

$$x_u \cdot x_v = \begin{cases} 90, & u = v, \\ 18, & u \sim v, \\ -6, & u \not\sim v. \end{cases}$$

Consequently, for $u \neq v$,

$$\|x_u - x_v\|^2 = \begin{cases} 144, & u \sim v, \\ 192, & u \not\sim v. \end{cases}$$

Proof. Let A be the adjacency matrix of Γ , and let J be the all-one matrix. Set

$$G = 96I + 24A - 6J.$$

Since Γ is strongly regular with parameters $(416, 100, 36, 20)$, its adjacency eigenvalues are

$$100, \quad 20, \quad -4$$

with multiplicities $1, 65, 350$. Hence G has eigenvalue

$$96 + 24 \cdot 100 - 6 \cdot 416 = 0$$

on the all-one line, eigenvalue $96 + 24 \cdot 20 = 576$ on the 65-dimensional 20-eigenspace, and eigenvalue $96 + 24(-4) = 0$ on the -4 -eigenspace. Hence G is positive semidefinite of rank 65, so it is a Gram matrix for vectors in $\mathbb{R}^{65}$. Its entries are exactly the stated inner products, and the distance formula follows from

$$\|x_u - x_v\|^2 = \|x_u\|^2 + \|x_v\|^2 - 2x_u \cdot x_v.$$

□

Only the input configuration $\{x_v : v \in V(\Gamma)\}$ is two-distance. The final set constructed below is not two-distance.

4 A codimension-two subspace

For $i = 1, 2, 3$, define

$$S_i = \sum_{y \in B_i} x_y.$$

Lemma 3. *The points x_c , $c \in C$, lie in a common 63-dimensional subspace of the standard $\mathbb{R}^{65}$ representation.*

Proof. Let $c \in C$. By Lemma 1, the vertex c has exactly 8 neighbors and 24 non-neighbors in each B_i . Therefore

$$x_c \cdot S_i = 8 \cdot 18 + 24 \cdot (-6) = 0.$$

Thus every x_c , $c \in C$, is orthogonal to S_1, S_2, S_3 .

The same degree data give

$$S_i \cdot S_i = 32(90 + 20 \cdot 18 + 11 \cdot (-6)) = 12288,$$

and, for $i \neq j$, since there are no edges between B_i and B_j ,

$$S_i \cdot S_j = 32^2(-6) = -6144.$$

Hence the Gram matrix of S_1, S_2, S_3 has diagonal entries 12288 and off-diagonal entries -6144 . Its eigenvalues are

$$0, \quad 18432, \quad 18432,$$

so $\dim \text{span}(S_1, S_2, S_3) = 2$.

Let

$$W = \text{span}(S_1, S_2, S_3)^\perp.$$

The ambient representation has dimension 65, so $\dim W = 63$. We have shown that $x_c \in W$ for every $c \in C$. $\square$

We identify W with $\mathbb{R}^{63}$ from now on.

5 The additional point

Lemma 4. *Choose $b \in B_1$, and define*

$$z_b = x_b - \frac{1}{32}S_1.$$

Then $z_b \in W$,

$$\|z_b\|^2 = 78,$$

and for every $c \in C$,

$$z_b \cdot x_c = \begin{cases} 18, & b \sim c, \\ -6, & b \not\sim c. \end{cases}$$

Proof. First,

$$x_b \cdot S_1 = 90 + 20 \cdot 18 + 11 \cdot (-6) = 384,$$

and

$$\frac{1}{32}S_1 \cdot S_1 = \frac{12288}{32} = 384.$$

Thus $z_b \cdot S_1 = 0$.

For $i = 2, 3$, the verified degree data say that there are no edges between B_1 and B_i . Hence

$$x_b \cdot S_i = 32(-6) = -192.$$

Also

$$\frac{1}{32}S_1 \cdot S_i = \frac{-6144}{32} = -192.$$

Thus $z_b \cdot S_i = 0$ for $i = 2, 3$. Therefore $z_b \in W$.

The norm is

$$\|z_b\|^2 = 90 - \frac{2}{32} \cdot 384 + \frac{1}{32^2} \cdot 12288 = 78.$$

Finally, Lemma 3 gives $S_1 \cdot x_c = 0$ for $c \in C$, so

$$z_b \cdot x_c = x_b \cdot x_c,$$

The displayed formula follows from Lemma 2. $\square$

Set

$$t = \frac{\sqrt{222} - 1}{13}, \quad p = tz_b.$$

Equivalently, t is the positive root of

$$78t^2 + 12t = 102.$$

Define

$$X = \{x_c : c \in C\} \cup \{p\}.$$

By Lemmas 3 and 4, all points of X lie in $W \simeq \mathbb{R}^{63}$. Also p is not one of the x_c : every x_c has squared norm 90, while $\|p\|^2 = 78t^2$; if $78t^2 = 90$, then the equation $78t^2 + 12t = 102$ would give $t = 1$, which is not a root of that equation. Hence

$$X \subset \mathbb{R}^{63}, \quad |X| = 321.$$

6 Diameter and clique bound

Lemma 5. *The set X has squared diameter 192. More precisely, for distinct $c, c' \in C$,*

$$\|x_c - x_{c'}\|^2 = \begin{cases} 144, & c \sim c', \\ 192, & c \not\sim c', \end{cases}$$

and for $c \in C$,

$$\|p - x_c\|^2 = \begin{cases} 192 - 48t, & b \sim c, \\ 192, & b \not\sim c. \end{cases}$$

Proof. The formula for distances between points $x_c, x_{c'}$ is Lemma 2. For distances from p , use $\|p\|^2 = 78t^2$ and Lemma 4. If $b \not\sim c$, then

$$\|p - x_c\|^2 = 78t^2 + 90 + 12t = 192.$$

If $b \sim c$, then

$$\|p - x_c\|^2 = 78t^2 + 90 - 36t = 192 - 48t < 192.$$

because $t > 0$.

Thus all squared distances in X are at most 192. The value 192 is attained between p and any $c \in C$ not adjacent to b ; such a c exists because b has 80 neighbors in C and $|C| = 320$. $\square$

Lemma 6. *Every subset $Y \subset X$ with*

$$\text{diam}(Y) < \text{diam}(X)$$

has at most 5 points.

Proof. First suppose that $p \notin Y$. Then $Y \subset \{x_c : c \in C\}$. By Lemma 5, two points $x_c, x_{c'}$ are at the full diameter exactly when $c \not\sim c'$. Hence the vertices corresponding to the points of Y must be pairwise adjacent. They form a clique in Γ , and Lemma 1 gives $|Y| \leq 5$.

Now suppose $p \in Y$. Any other point of Y is some x_c . In order to have distance strictly smaller than $\text{diam}(X)$ from p , Lemma 5 forces $b \sim c$. Also, the remaining vertices c must be pairwise adjacent. Therefore

$$\{b\} \cup \{c \in C : x_c \in Y \setminus \{p\}\}$$

is a clique in Γ . By Lemma 1, this clique has at most 5 vertices. Since Y is obtained from this clique by replacing b with p , we get $|Y| \leq 5$. $\square$

Proof of Theorem 1. The set X has 321 points in $\mathbb{R}^{63}$. By Lemma 6, every subset of X of strictly smaller diameter has at most 5 points. Thus any partition of X into subsets of smaller diameter requires at least

$$\left\lceil \frac{321}{5} \right\rceil = 65$$

parts. Since $65 > 64 = 63 + 1$, Borsuk's conjecture fails in dimension 63. $\square$

7 Computational verification

The finite checks in Lemma 1 are reproduced by the verification script in the accompanying repository [4]. The script reconstructs the graph from $\text{PG}(2, 16)$, verifies the strongly regular parameters $(416, 100, 36, 20)$, builds the partition B_1, B_2, B_3, C , checks the degree data used in Lemma 3, and checks the clique obstruction used in Lemma 6. It also checks that the selected vertex b has no 5-clique inside $N(b) \cap C$; this last check is a consistency check and is not needed for the proof above.

The script is deterministic and uses exact finite-field arithmetic and integer bitsets.

Statement on AI assistance

The construction and proof were obtained with assistance from GPT-5.5 Pro.

References

- [1] A. V. Bondarenko, *On Borsuk's conjecture for two-distance sets*, Discrete & Computational Geometry **51** (2014), 509–515. <https://arxiv.org/abs/1305.2584>
- [2] T. Jenrich and A. E. Brouwer, *A 64-dimensional counterexample to Borsuk's conjecture*, Electronic Journal of Combinatorics **21** (2014), no. 4, Paper 4.29. <https://www.combinatorics.org/ojs/index.php/eljc/article/view/v21i4p29/pdf>
- [3] A. E. Brouwer, *The $G_2(4)$ graph*, graph data and construction notes. <https://aeb.win.tue.nl/graphs/G24.html>
- [4] M. Grinsztajn, *A 63-dimensional counterexample to Borsuk's conjecture*, accompanying repository. <https://github.com/maaxgrin/borsuk-63-counterexample>